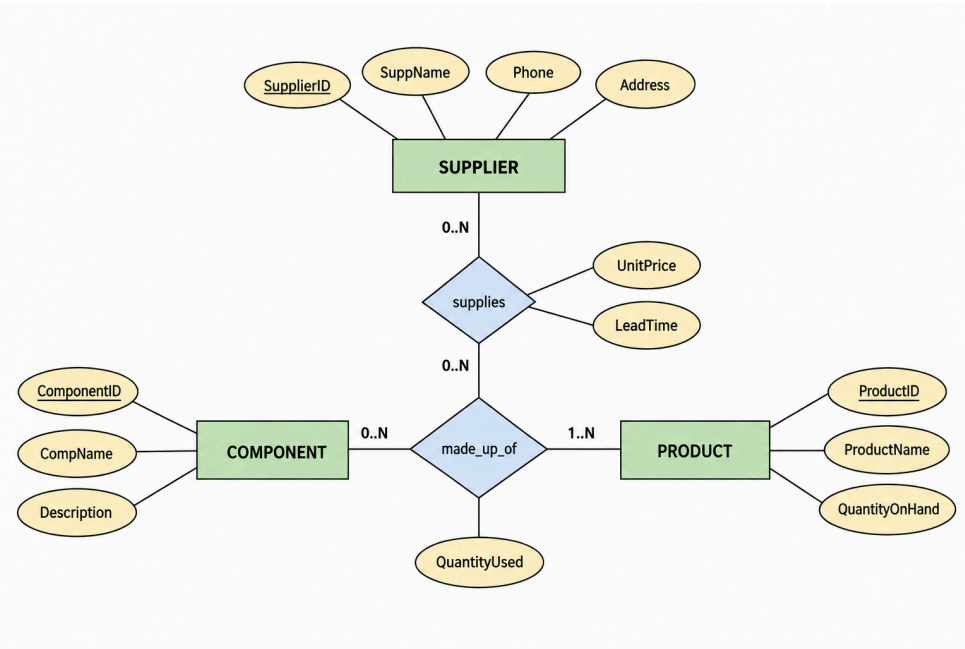


DBMS LAB ASSIGNMENT - 3

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Q1 — Assumptions

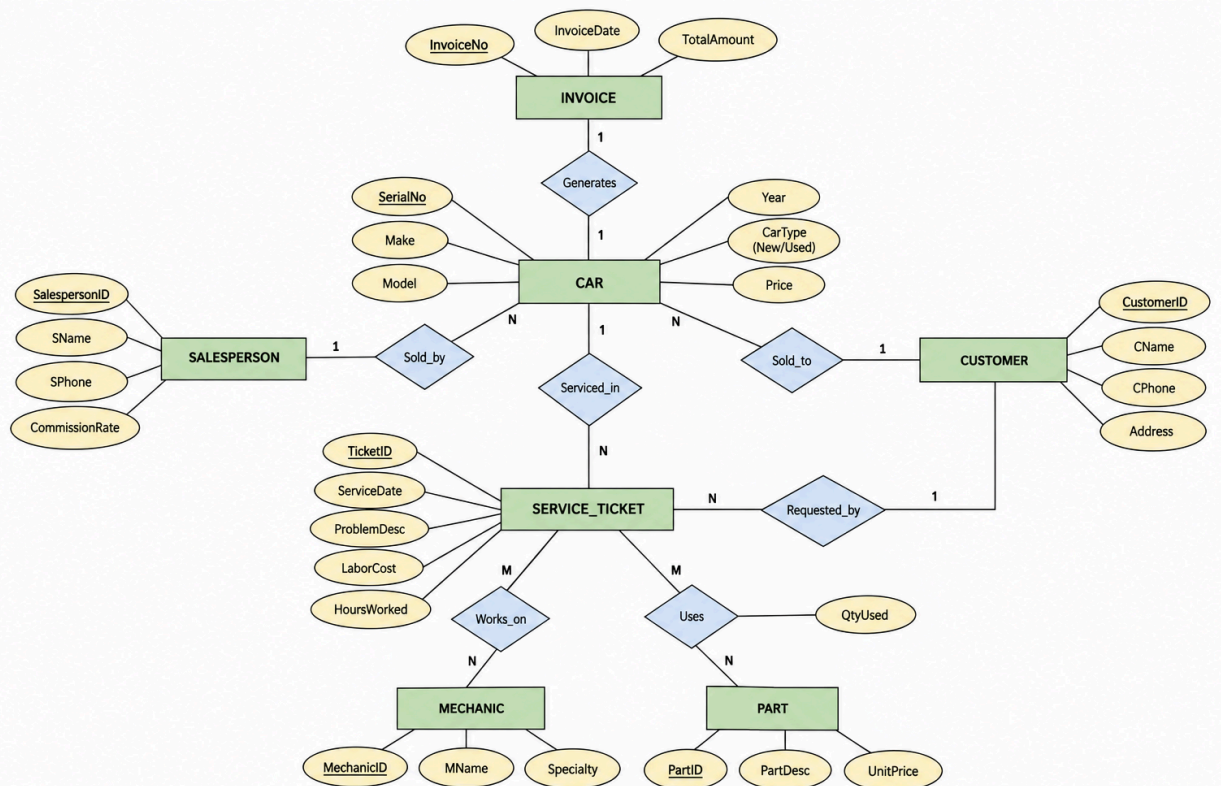
1. Each Product is uniquely identified by ProductID.
2. Each Component is uniquely identified by ComponentID.
3. Each Supplier is uniquely identified by SupplierID.
4. A product must contain at least one component, but a component may exist without being used in any product.
5. A component may be supplied by zero or more suppliers, and a supplier may supply zero or more components.
6. The combination of ProductID and ComponentID uniquely identifies a product-component relationship.
7. **QuantityUsed** represents the quantity of a particular component required for a particular product.
8. The combination of SupplierID and ComponentID uniquely identifies a supplier-component relationship.
9. **UnitPrice** and **LeadTimeDays** are attributes of the supplier–component relationship, since the price and lead time may differ between suppliers.
10. A supplier can exist without supplying any component.
11. Component and product IDs are assumed to be non-null and unique.



PRODUCT (<u>ProductID</u> , ProductName, QuantityOnHand) PK: ProductID	COMPONENT (<u>ComponentID</u> , ComponentName, Description) PK: ComponentID
SUPPLIER (<u>SupplierID</u> , SupplierName, ContactPhone, Address) PK: SupplierID	PRODUCT_COMPONENT (<u>ProductID</u> , <u>ComponentID</u> , QuantityUsed) PK: (<u>ProductID</u> , <u>ComponentID</u>) FK: ProductID → PRODUCT (ProductID) FK: ComponentID → COMPONENT (ComponentID)
SUPPLIER_COMPONENT (<u>SupplierID</u> , <u>ComponentID</u> , UnitPrice, LeadTimeDays) PK: (<u>SupplierID</u> , <u>ComponentID</u>) FK: SupplierID → SUPPLIER (SupplierID) FK: ComponentID → COMPONENT (ComponentID)	

Q2 — Assumptions

1. Each car is uniquely identified by **SerialNo**.
2. A car may initially exist as unsold inventory, so **SalespersonID** and **CustomerID** can be NULL until the car is sold.
3. Once a car is sold, it has exactly one salesperson.
4. Once a car is sold, it has exactly one customer.
5. A salesperson can sell zero, one, or many cars.
6. A customer can purchase zero, one, or many cars.
7. Each car sale generates exactly one invoice.
8. Each invoice belongs to exactly one car sale, so **SerialNo** in **INVOICE** is NOT NULL and UNIQUE.
9. A customer can use the dealership's service facility without purchasing a car.
10. Each service visit creates a separate Service Ticket, uniquely identified by **TicketID**.
11. Every service ticket is associated with exactly one car, identified by **SerialNo**.
12. A car can have zero, one, or many service tickets, allowing its complete service history to be maintained.
13. A service ticket may involve zero, one, or many mechanics, and a mechanic may work on zero, one, or many service tickets.
14. The combination of TicketID and MechanicID uniquely identifies a **WORKS_ON** record.
15. A service ticket may use zero, one, or many replacement parts, while a part may be used on many service tickets.
16. The combination of TicketID and PartID uniquely identifies a **USES** record.
17. **HoursWorked** is an attribute of the service-ticket–mechanic relationship.
18. **QtyUsed** is an attribute of the service-ticket–part relationship.



SALESPERSON (<u>SalespersonID</u> , SName, SPhone, CommissionRate) PK: SalespersonID	CUSTOMER (<u>CustomerID</u> , CName, CPhone, CAddress) PK: CustomerID
MECHANIC (<u>MechanicID</u> , MName, Specialty) PK: MechanicID	PART (<u>PartID</u> , PartDesc, UnitPrice) PK: PartID
CAR (<u>SerialNo</u> , Make, Model, Year, CarType, Price, SalespersonID, CustomerID) PK: SerialNo FK: SalespersonID → SALESPERSON (SalespersonID) [nullable — unsold car] FK: CustomerID → CUSTOMER (CustomerID) [nullable — unsold car]	INVOICE (<u>InvoiceNo</u> , InvoiceDate, TotalAmount, SerialNo) PK: InvoiceNo FK: SerialNo → CAR (SerialNo) [not null, unique — one invoice per car sale]
SERVICE_TICKET (<u>TicketID</u> , ServiceDate, ProblemDesc, LaborCost, SerialNo, CustomerID) PK: TicketID FK: SerialNo → CAR (SerialNo) FK: CustomerID → CUSTOMER (CustomerID)	WORKS_ON (<u>TicketID</u> , <u>MechanicID</u> , HoursWorked) PK: (TicketID, MechanicID) FK: TicketID → SERVICE_TICKET (TicketID) FK: MechanicID → MECHANIC (MechanicID)
USES (<u>TicketID</u> , <u>PartID</u> , QtyUsed) PK: (TicketID, PartID) FK: TicketID → SERVICE_TICKET (TicketID) FK: PartID → PART (PartID)	